

POKHARA UNIVERSITY

Level: Bachelor

Semester: Spring

Year: 2025

Programme: BCSIT

Full Marks: 100

Course: Object Oriented Language (Java)

Pass Marks: 45

Time: 3 hrs.

Candidates are required to answer in their own words as far as practicable. The figures in the margin indicate full marks.

Section "A"

Very Short Answer Questions

Attempt all the questions. $[10 \times 2 = 20]$

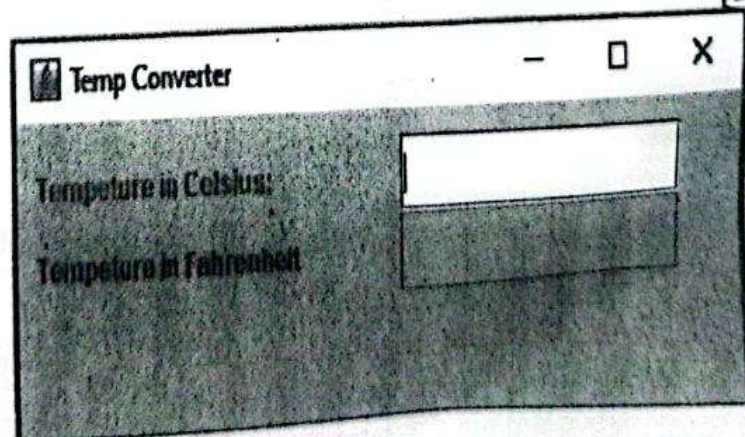
1. Define class and object.
2. Justify the statement "Java is platform independent".
3. Explain Garbage collection in java.
4. Define JDK, JRE and JVM.
5. What is difference between Label and JLabel class?
6. State the syntax to import a specific class from a package.
7. Give two examples of exceptions in Java.
8. What is a generic class in Java?
9. What are layout managers in Swing? Name two types & their usage.
10. Differentiate between input streams and output streams in Java. Provide one example each.

Section "B"

Descriptive Answer Questions

Attempt any five questions. $[5 \times 10 = 50]$

11. Define polymorphism. Differentiate between compile time polymorphism and runtime polymorphism with example of each.
12. Differentiate between checked and unchecked exceptions with examples.
13. What is the purpose of **this** keyword in Java? Explain different types of constructors along with example.
14. a. Write a program to design the following UI using swing components. [5]



- b. When user enters temperature in Celsius and press enter key you must convert temperature in Celsius into Fahrenheit display through the respective text field as shown in the figure above. [Let $1^{\circ}\text{C} = 32^{\circ}\text{F}$]
[5]

15. What is **FileInputStream** and **FileOutputStream** in Java? Give examples.
16. Write a program to read 10 integer number from user and store in one dimension array and find the largest number of an array and display.

Group "C"

Long Answer Questions

Attempt any two questions. [2×15=30]

17. Write a complete Java program using Swing that accepts two numbers and perform addition, subtraction, multiplication and division and display the result.
18. a. Explain the purpose of "this" pointer in Java. Also include an example.
b. Explain the purpose and syntax of the try, catch, and finally blocks.
19. a. Write a generic Java program to store and display values of different data-types using a generic class and method.
b. Write a java program to calculate simple interest.